E-Commerce Sales Analysis Report

Phases 1–4 | Data Cleaning · EDA · Charts · Business Decisions

Dataset: UCI Online Retail | 541,909 rows | Dec 2010 – Dec 2011

Source: kaggle.com/datasets/carrie1/ecommerce-data

Prepared by Automated Analysis Pipeline

Executive Summary

This report covers a full four-phase analysis of 541,909 raw transactions from a UK-based e-commerce retailer (December 2010 – December 2011). After cleaning, **524,878** valid transactions remained, generating **£10.64M** in revenue across **19,960** orders from **38** countries.

£10.64 M	19,960	4,338	£533	4,026	38	£303	84.6%
Total Revenue	Total Orders	Unique Customers	Avg Order Value	Unique Products	Countries	Median Order	UK Revenue Share

Key Findings

- The United Kingdom accounts for 84.6% of revenue — significant concentration risk.
- Revenue peaks in November 2011 at £1.50M, over 2× the January baseline.
- 75% of transactions occur between 10:00–15:00; Sunday is the quietest day.
- Netherlands, EIRE, Germany and France are the strongest international expansion targets.
- 17,031 rows removed during cleaning (3.2% of raw data).

Phase 1 — Data Cleaning

Systematic data quality issues were identified and resolved before analysis.

Issue	Rows Affected	Action	Reason
Missing CustomerID	135,080	Flag as "Guest"	Guest revenue is valid
Missing Description	1,454	Drop row	Unidentifiable product
Duplicate rows	5,268	Remove	Inflates all metrics
Cancellation invoices	9,288	Remove	Returns, not sales
Negative Quantity	10,624	Remove	Return quantities
Negative UnitPrice	2,517	Remove	Pricing data errors

Data Type Corrections

InvoiceDate	String → datetime using format %m/%d/%Y %H:%M
CustomerID	Numeric float → string; nulls replaced with "Guest"
InvoiceNo	Cast to string to preserve codes and detect cancellation prefix "C"

Derived Columns Created

Revenue	Quantity × UnitPrice
YearMonth	floor_date(Date, "month") — for time-series aggregation
Hour	hour(Date) — intraday demand analysis
DayOfWeek	Day name — weekly pattern analysis
Strategy	Percentile rank classification: PROMOTE Star / High-Value / Monitor / DISCONTINUE

Final clean dataset: **524,878 rows** (reduced from 541,909, -17,031 rows / -3.2%). Exported as cleaned_dataset.csv.

Phase 2 — Exploratory Data Analysis

2.1 Unique Customers and Products

Registered customers	4,338
Guest transactions	135,080 rows without CustomerID
Unique products	4,026 distinct descriptions
Countries	38 in the cleaned dataset

2.2 Distribution of Quantities and Prices

Quantities are heavily right-skewed — most lines are 1–12 units with a long tail of bulk B2B orders. Unit prices mostly fall between £0.50 and £10.00; a small number of high-value items and admin lines (shipping, manuals) exceed £100.

2.3 Country Frequency

The UK dominates at **84.6%** of revenue (£9.00M). Outside the UK, Germany, France, EIRE and Netherlands have the highest transaction counts. This concentration is both a strength and a single-market risk.

2.4 Price Outliers

Items above £100 are predominantly service/postage lines (e.g., DOTCOM POSTAGE at up to £950). These should be excluded from product ranking analysis. Items below £0.10 are typically samples or damaged-goods write-offs.

2.5 Average Order Value

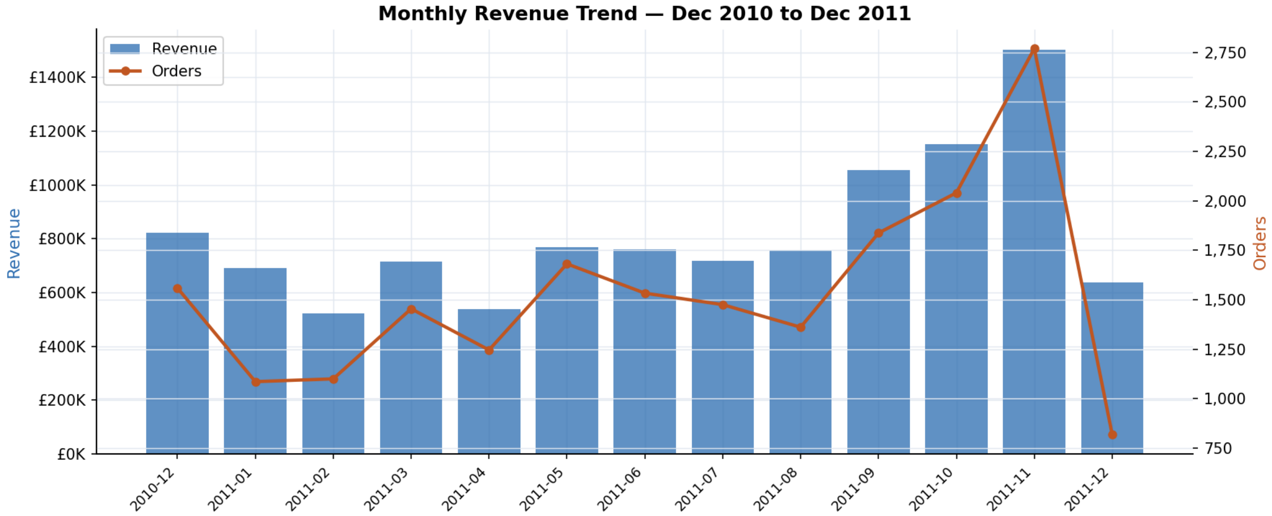
Mean order value	£533.17 — inflated by large wholesale orders
Median order value	£303.30 — better represents the typical transaction

2.6 Peak Time Periods

Approximately 75% of transactions occur between **10:00 and 15:00**. Tuesday–Thursday are the busiest days; Sunday is the quietest. November 2011 is the peak month (£1.50M) — over 2× January's £0.69M.

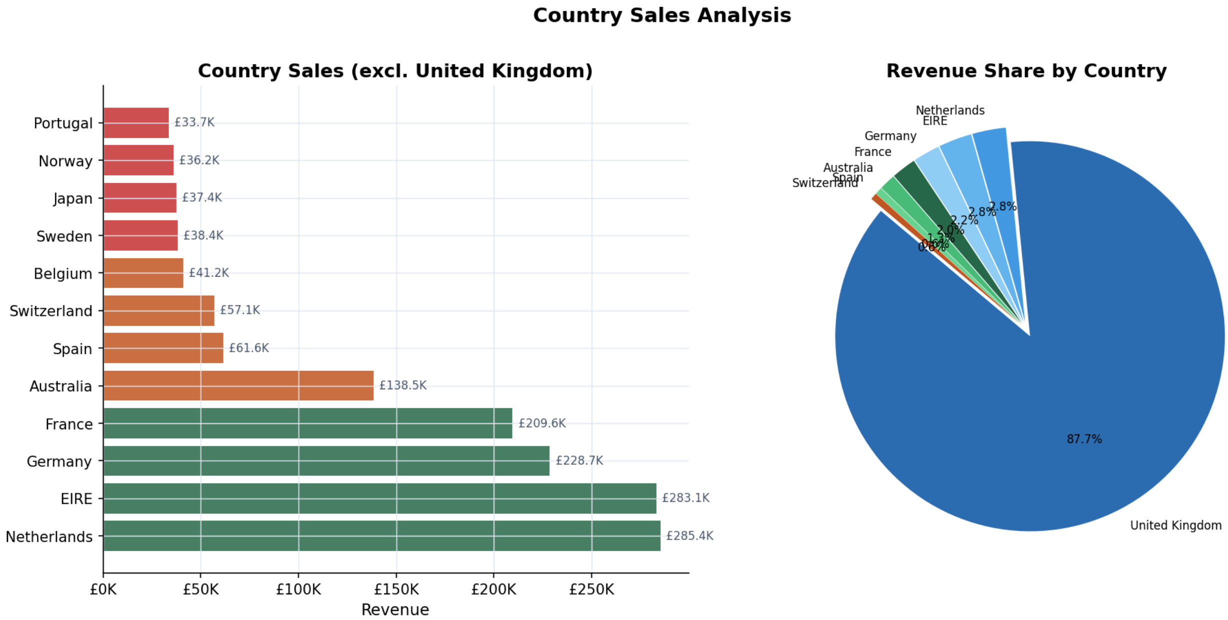
Phase 3 — Graphical Reports

Chart 1 — Revenue Trend Over Time



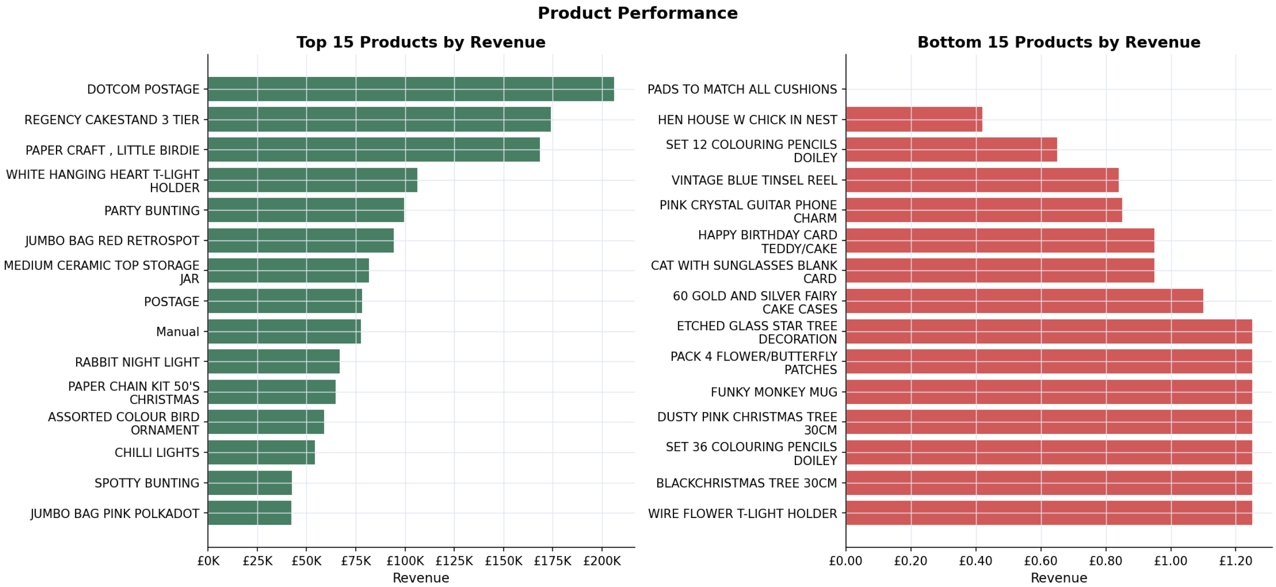
Monthly revenue (bars) and order volume (line). Revenue accelerates from September, peaking at £1.50M in November 2011.

Chart 2 — Country Sales Comparison



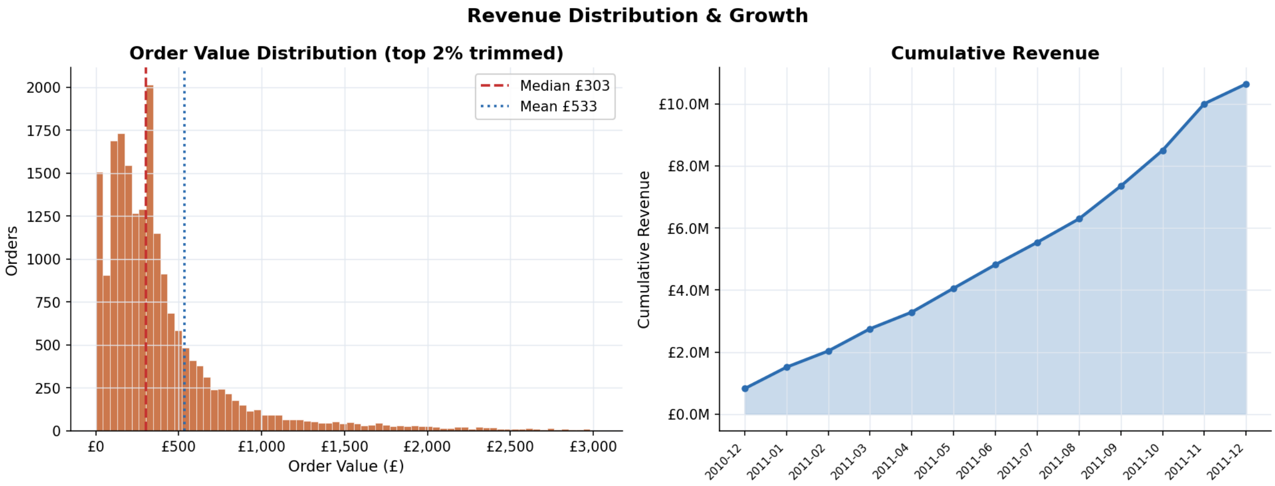
Left: Top international markets (excl. UK) coloured by priority tier. Right: Revenue share including UK (84.6%).

Chart 3 — Product Performance



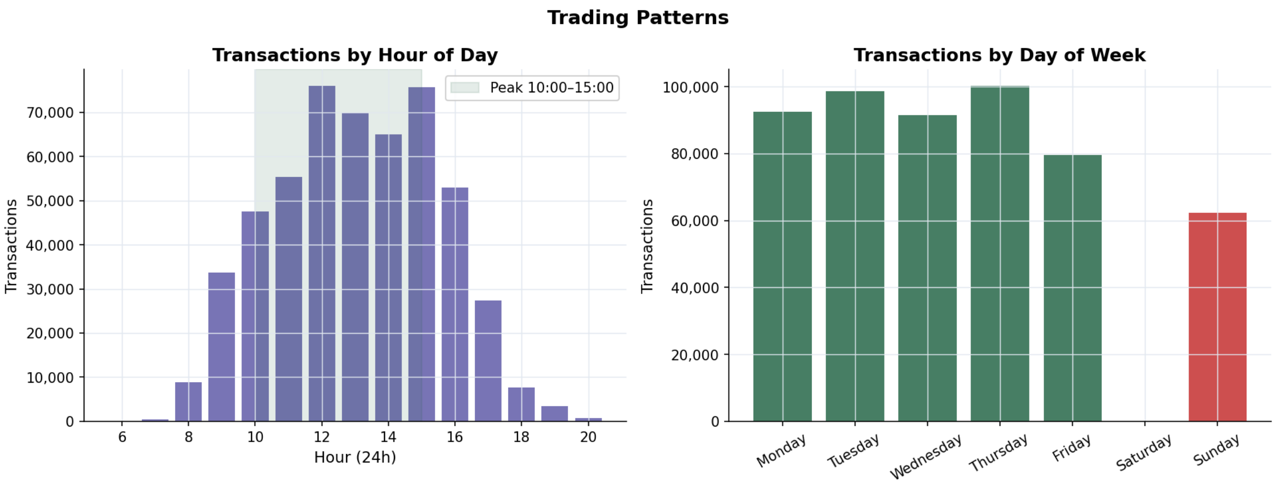
Top 15 products (green) and bottom 15 products (red) by total revenue.

Chart 4 — Order Value Distribution & Cumulative Revenue



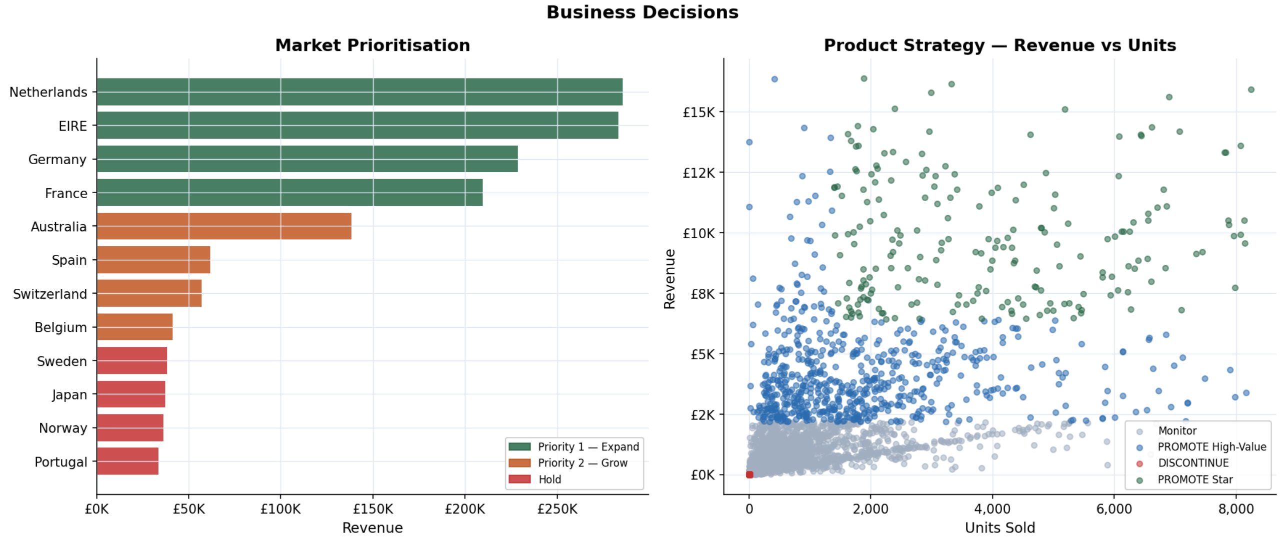
Left: Order value histogram with median/mean lines. Right: Cumulative revenue curve.

Chart 5 — Trading Patterns



Left: Transaction volume by hour — peak window 10:00–15:00 in green. Right: Transactions by day of week.

Chart 6 — Business Decision Visuals



Left: Market prioritisation tiers. Right: Product strategy scatter by revenue vs units.

Phase 4 — Business Decisions

4.1 Market Prioritisation

With 84.6% of revenue from the UK, reducing market concentration is the highest-impact strategic lever. Three-tier prioritisation for international markets:

Priority	Markets	Rationale
PRIORITY 1	Netherlands, EIRE, Germany, France	Highest international revenue. Invest in localised campaigns and dedicated a
PRIORITY 2	Australia, Switzerland, Belgium, Sweden	Mid-tier with healthy order values. Low-cost digital (email, social) sufficient.
HOLD	All others < £5K	Serve reactively. No dedicated budget until revenue threshold is met.

4.2 Product Strategy — Promote vs Discontinue

Classification	Criteria	Action
PROMOTE — Star	Top 10% revenue AND top 25% units	Bundle deals, homepage placement, priority campaigns. Ensure stock depth
PROMOTE — High-Value	Top 25% revenue	Premium positioning. Protect margin; avoid deep discounting.
Monitor	Mid-tier	Flag after 2 consecutive months of decline.
DISCONTINUE	Bottom 10% revenue AND bottom 10% units	Clear at discount. Remove from catalogue. Reallocate warehouse space.

4.3 Stock Level Planning

Restock trigger	Place bulk orders June/July; build warehouse stock from August.
Sep–Nov uplift	Revenue surges ~39% above average in Sep, peaking at £1.50M in November.
December caution	Revenue drops post-Christmas. Keep Dec/Jan inventory lean.
Peak hours	10:00–15:00 accounts for 75% of orders. Full warehouse staffing by 09:30.
Slow season	January–February: stocktake, discontinue-SKU clearance, supplier renegotiation.

Appendix — Dataset & Methodology

Dataset Information

Source	Kaggle — carrie1/ecommerce-data (UCI Online Retail)
Raw rows	541,909
Clean rows	524,878
Date range	December 2010 – December 2011
Columns	InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country
File encoding	latin-1 (ISO-8859-1)

Tools Used

Dashboard	R — shiny, shinydashboard, plotly, DT, dplyr, lubridate, scales
Analysis	Python — pandas, numpy, matplotlib
Report	Python — reportlab

Key Methodological Notes

· CustomerID nulls were retained as "Guest" (not dropped) to avoid removing 25% of revenue. · Cancellation invoices (prefix "C") represent refunds and are excluded from all sales metrics. · Service line items (DOTCOM POSTAGE, MANUAL) are excluded from product promotion decisions even if they rank highly by revenue. · Product Strategy percentile thresholds: Star = rev ≥ p90 AND units ≥ p75; High-Value = rev ≥ p75; Discontinue = rev ≤ p10 AND units ≤ p10.